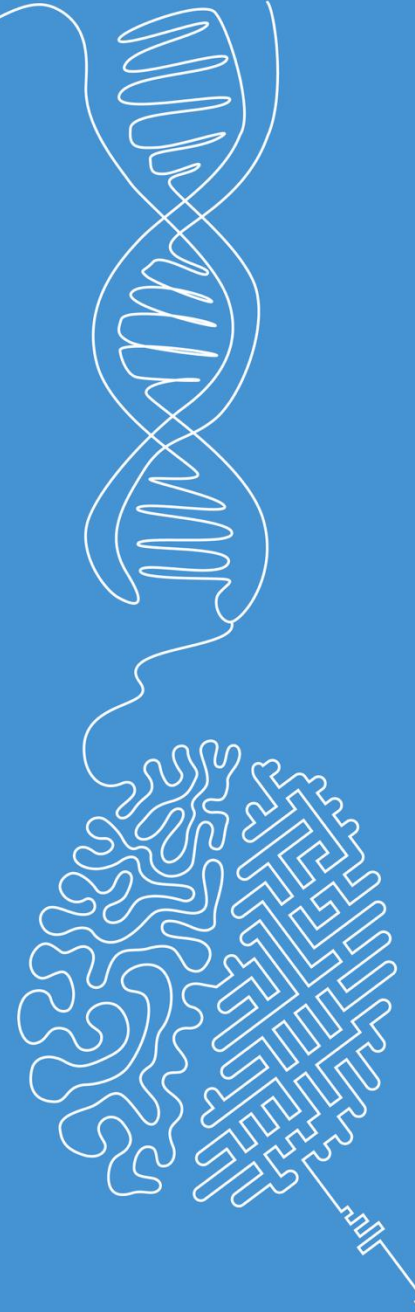


Image transformations

Image and Signal Processing

Norman Juchler



Motivation

- Image transformations change how an image is represented in space, without altering the underlying scene content.
- Where we encounter image transformations:
 - Alignment and registration
→ Match images from different times, sensors, or viewpoints
 - Distortion correction
→ Remove perspective effects or camera-induced artifacts
 - Resizing and normalization
→ Adapt images to required resolution or formats
 - Data augmentation (machine learning)
→ Generate additional training data (e.g., via rotations, flips, scaling)
 - Artistic effects and image manipulation
→ Create visual effects, warping, or stylized transformations

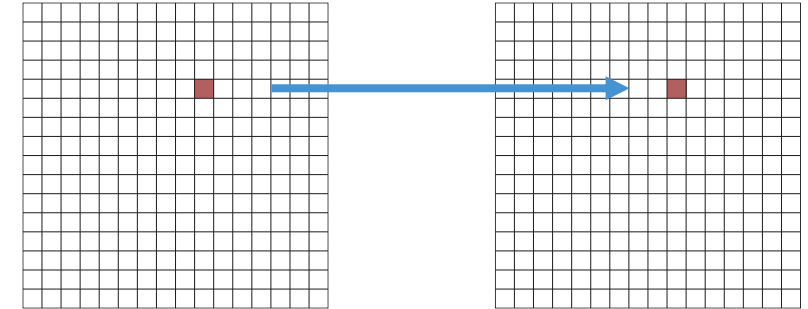


Source: fotor.com

Classification of transformations

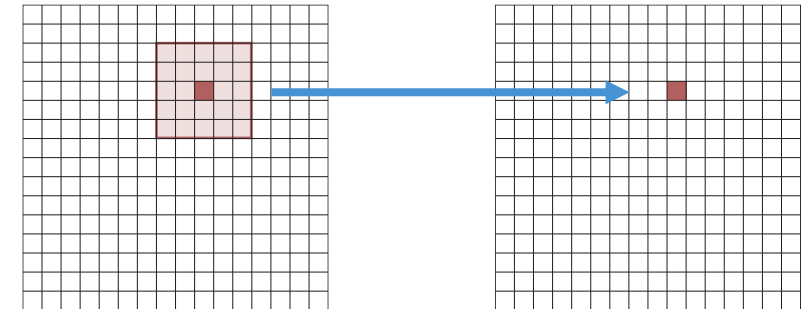
■ Pixel transformations (point operations)

- Operate on each pixel independently
- Modify pixels without considering spatial context
- Examples:
 - Thresholding
 - Brightness / contrast adjustment
 - Color transformations



■ Patch-based transformations (neighborhood operations)

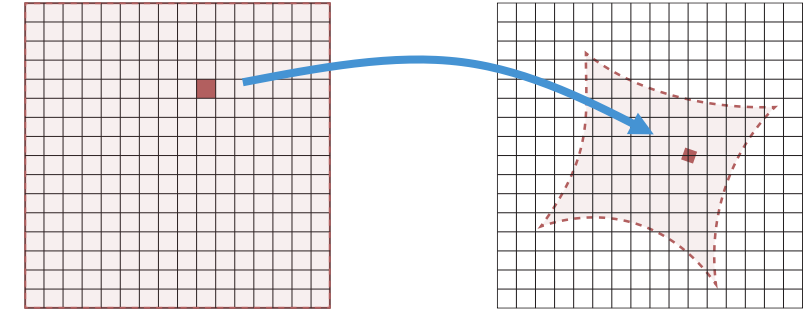
- Operate on a pixel using its local neighborhood
- Capture local structure and spatial context
- Examples:
 - Gaussian blur
 - Edge filter



Classification of transformations

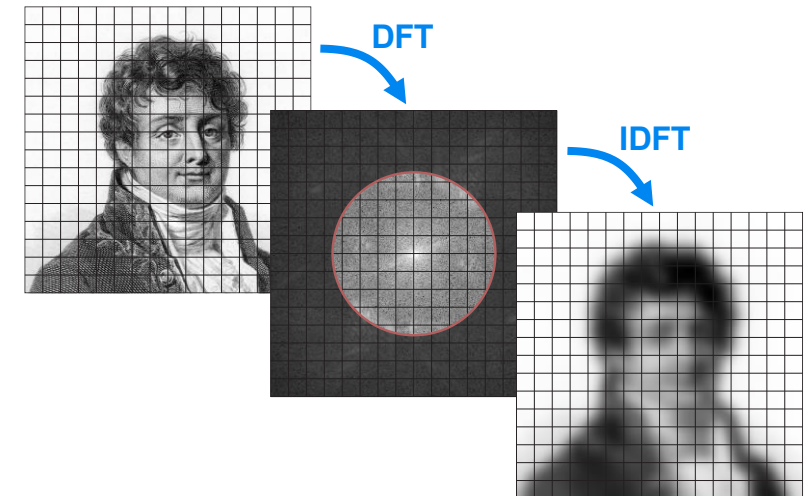
■ Geometric transformations

- Modify the spatial position of pixels
- Map coordinates: $(x, y) \rightarrow (x', y')$
- Examples:
 - Affine transformations (translation, rotation, scaling)
 - Projective (perspective) transformations
 - Non-linear transformation (warping)



■ Frequency domain transformations

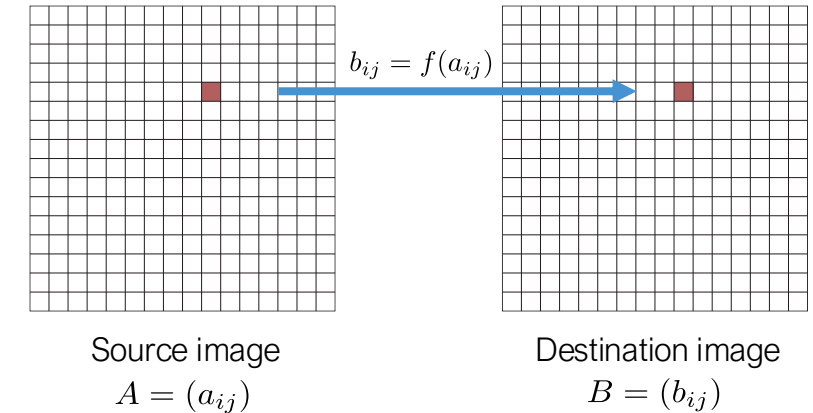
- Operate on the image in the **frequency domain** instead of spatial domain
- Modify frequency components (e.g., remove noise)
- Examples:
 - Low-pass filtering
 - High-pass filtering



Point operations

Definition

- In a point operation, the pixel value of the destination image B at location (i,j) is computed solely from the pixel value(s) at the same location (i,j) in the source image(s) A .
- Point operations process each pixel independently, without using information from nearby pixels.



Classification:

- Homogeneous vs. inhomogeneous operations
 - Homogeneous: Same operation applies to all pixels (i,j)
 - Inhomogeneous: Operation depends on pixel location (i,j)
- m-input operations: Number of input values used per pixel (e.g., RGB $\rightarrow$ grayscale)
- n-valued operations: Number of output values produced per pixel (e.g., grayscale $\rightarrow$ RGB, feature maps)

The following slides illustrate different examples of point operations.

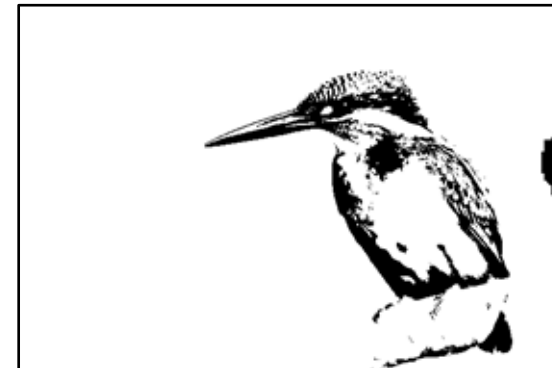
Thresholding

- Thresholding converts a grayscale image into a binary image by comparing each pixel value to a threshold.

- For a grayscale image A and a threshold T:
$$b_{ij} = \begin{cases} 1 & a_{ij} \geq T \\ 0 & a_{ij} < T \end{cases}$$



T = 130



Grayscale conversion

- Each pixel in a color image can be converted into a single gray value.
- **Example:** Weighted intensity (luminance)

$$I(i, j) = 0.299 \cdot R(i, j) + 0.587 \cdot G(i, j) + 0.114 \cdot B(i, j)$$

- Combines the red, green, and blue channels
- Weights reflect human sensitivity to different colors
- Produces a perceptually meaningful grayscale image



Grayscale conversion

- Each pixel in a color image can be converted into a single gray value.
- Alternative** grayscale conversions

Input image



value



$$V = \max(R, G, B)$$

luminance



$$L = \frac{\max(R, G, B) + \min(R, G, B)}{2}$$

intensity



$$I = \frac{R + G + B}{3}$$

weighted intensity



$$I = 0.3 \cdot R + 0.6 \cdot G + 0.1 \cdot B$$

Pseudo-color / color mapping

- Grayscale images can be converted into **pseudo-color images** using a **color lookup table (LUT)**:

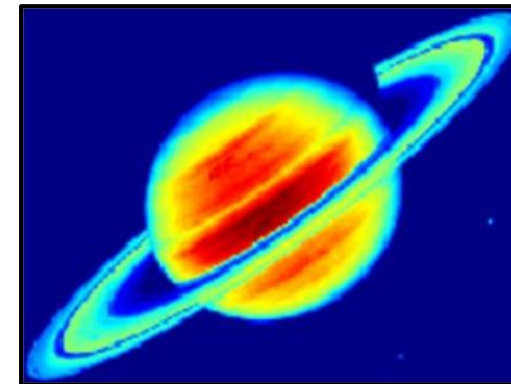
- Maps each gray value to a color (RGB)
- Has size $k \times 3$ where k is the number of gray levels
- Each entry defines an (R,G,B) value

$$T_{\text{color}} = \begin{bmatrix} r_0 & g_0 & b_0 \\ r_1 & g_1 & b_1 \\ r_2 & g_2 & b_2 \\ \vdots & \vdots & \vdots \\ r_{255} & g_{255} & b_{255} \end{bmatrix}$$



A lookup table (LUT) can be represented as a matrix in which each row i assigns a grayscale value $v=i$ to a corresponding color (R,G,B).

- **Example:**



Arithmetic operations on images

- Arithmetic operations combine **two images** by applying pixel-wise operations to their intensity (or color) values: $A, B \rightarrow C$
- Example:**

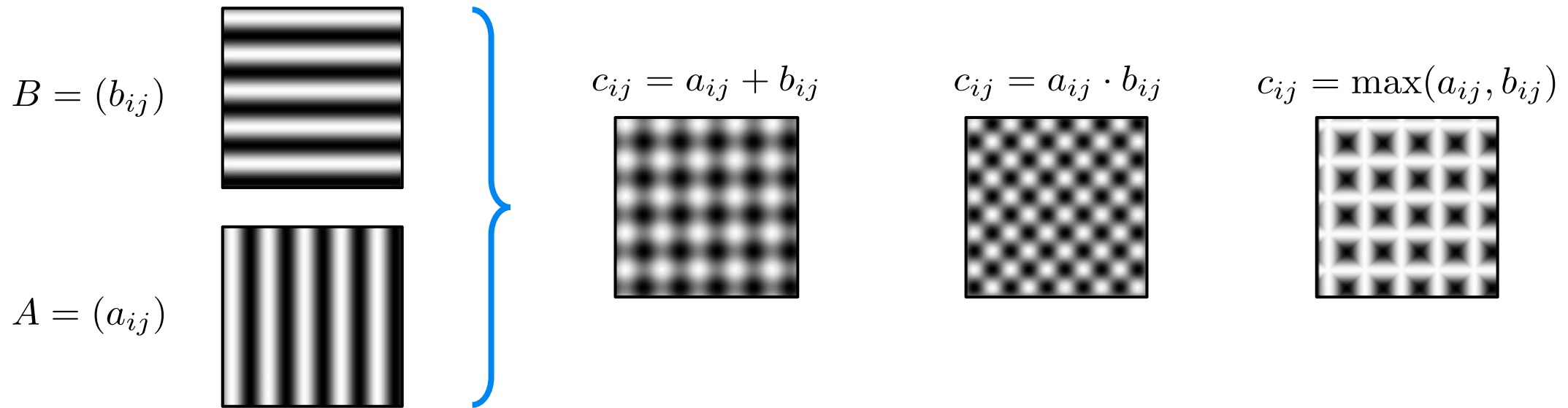


Image blending

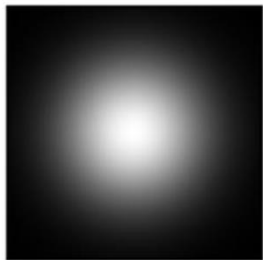
- Two images can be blended either homogeneously or inhomogeneously.
 - Homogeneous blending → The same weighting is applied to all pixels
 - Inhomogeneous blending → The weighting varies across the image (e.g., spatial masks)



$\mathbf{A} = (a_{ij})$



$\mathbf{B} = (b_{ij})$



mask for inhom.
blending



homogeneous blending

$$c_{ij} = \gamma a_{ij} + (1 - \gamma) b_{ij}$$

$$\gamma \in [0, 1]$$



inhomogeneous blending

$$c_{ij} = \gamma_{ij} a_{ij} + (1 - \gamma_{ij}) b_{ij}$$

$$\gamma_{ij} \in [0, 1] \quad \mathbf{M} = (\gamma_{ij})$$

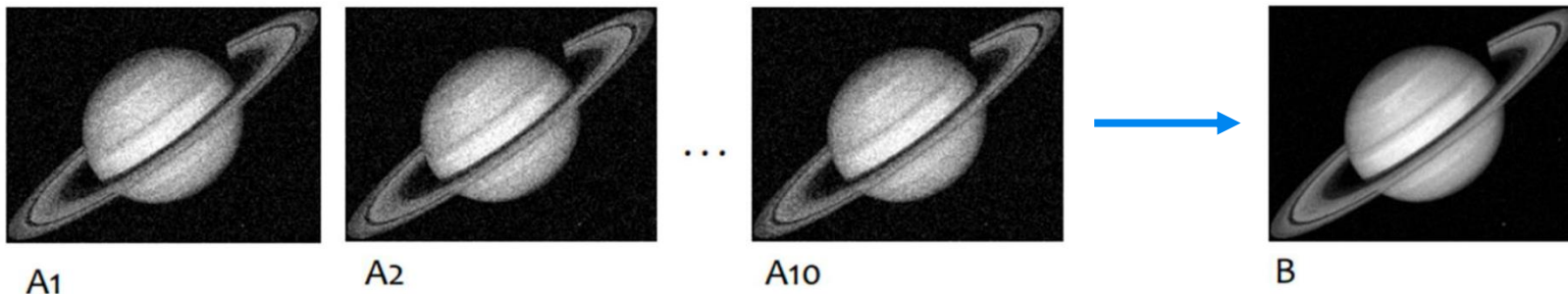
Example: Averaging

- From statistics: The mean of n samples x_i approximates the expected value of a random variable X :

$$\bar{x} = \frac{1}{n} (x_1 + x_2 + \dots + x_n) \xrightarrow{n \rightarrow \infty} E(X)$$

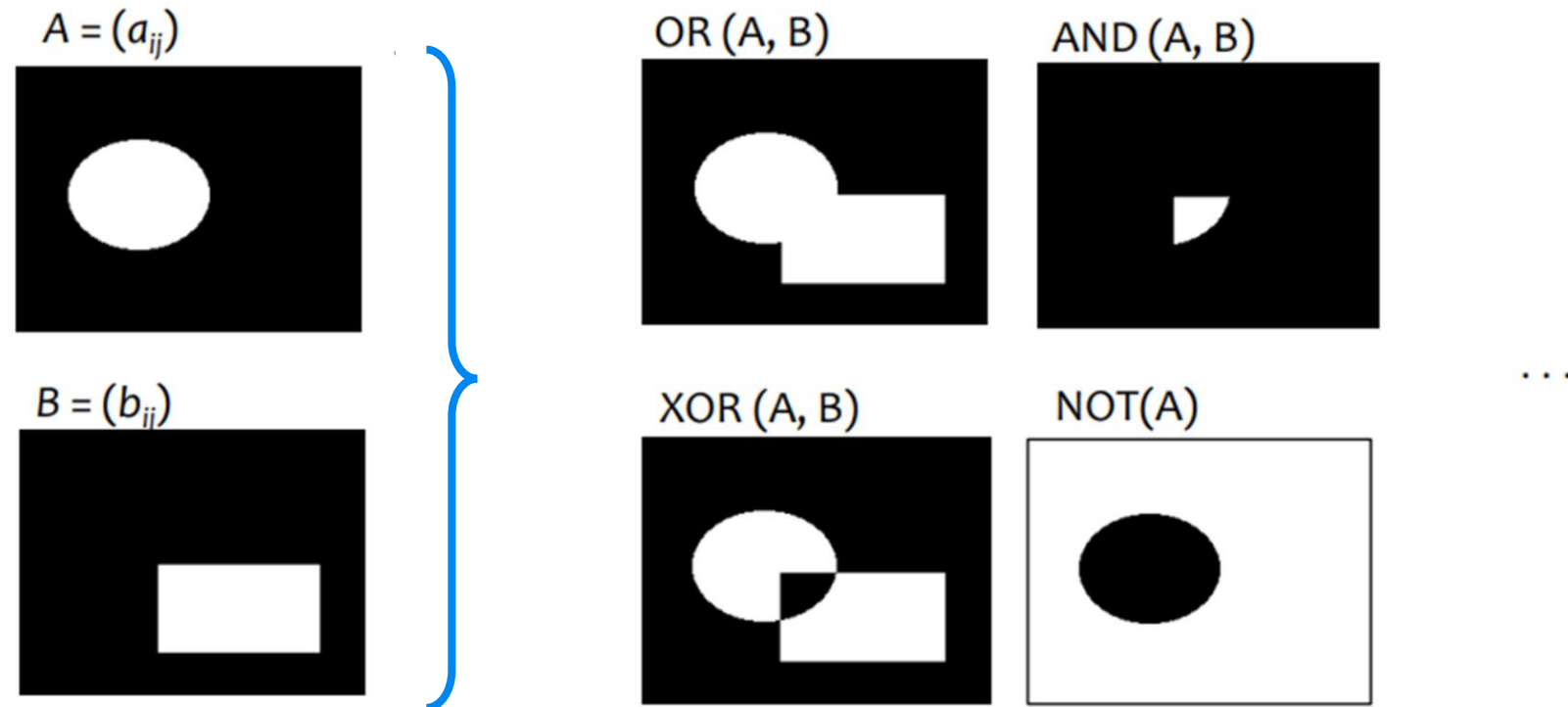
- Application to images: If multiple noisy observations of the same image are available, random acquisition noise can be reduced by averaging them.

$$f(A_1, A_2 \dots A_n) \rightarrow B : \quad B(i, j) = \frac{1}{n} \sum_{k=1}^n A_k(i, j)$$



Logical operations

- Logical operations are defined for binary images, where pixel values represent logical states (e.g., 0 and 1).
- They combine or modify images using Boolean rules applied pixel-wise.



Intensity transformations

- Modify the range or distribution of intensity (gray) values in an image.
- Purpose:
 - Enhance contrast
 - Improve visibility of structures
 - Adapt intensity range for further processing



Example: Contrast enhancement results using different methods. Each method analyzes the distribution of pixel intensities (globally or locally on patches) and applies a mapping to improve contrast. Methods shown: Intensity remapping (map 1st and 99th percentile to black and white), histogram equalization, and contrast-limited adaptive histogram equalization (CLAHE). Source: [MATLAB](#)

Intensity transformations: Intensity mapping

- How are intensity transformations applied?

- Approach 1: Transfer functions**

- Also known as intensity mapping
- Define a function that maps input intensities to output values:

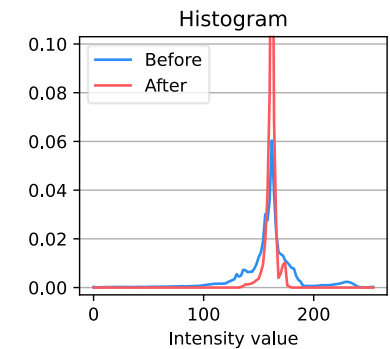
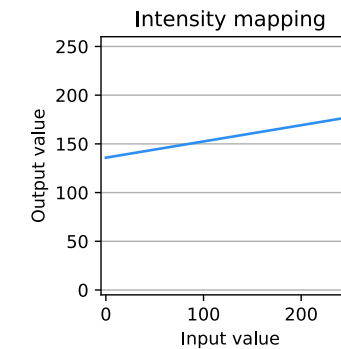
$$B(i, j) = f(A(i, j))$$

- Examples:
 - Linear scaling (contrast stretching)
 - Gamma correction
 - Thresholding

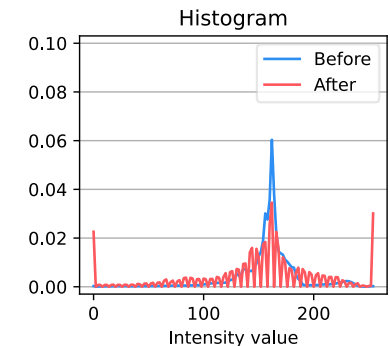
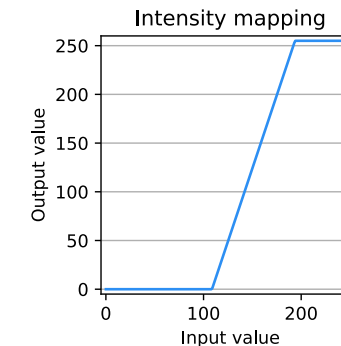


Example: A grayscale image (left) transformed using different linear intensity mappings, which define how input intensities (x-axis) are mapped to output values (y-axis). The histograms show the intensity distributions before and after the transformation.

Contrast squeeze



Contrast stretch



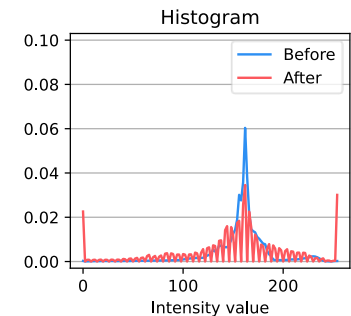
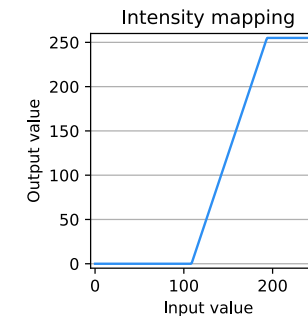
Intensity transformations: Intensity mapping

- Various functional relationships can be defined between input and output intensity values:

- **Linear function:**

- Maps intensity values linearly to a new range
- Simple and widely used for contrast adjustment
- Function that maps the intensity range of the input image $A(i, j) \in [A_{\min}, A_{\max}]$ (data) to a new value range $B(i, j) \in [B_{\min}, B_{\max}]$ (parameters).

$$B(i, j) = B_{\min} + \frac{A(i, j) - A_{\min}}{A_{\max} - A_{\min}} (B_{\max} - B_{\min})$$



Intensity transformations: Intensity mapping

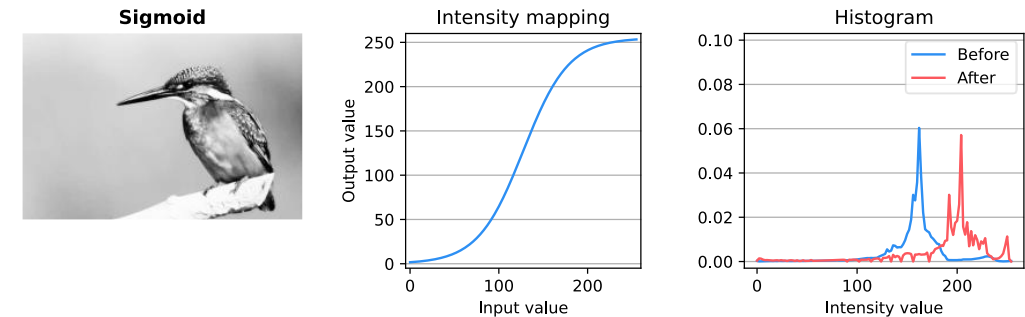
- Various functional relationships can be defined between input and output intensity values:

- **Sigmoid function:**

- Maps intensity values to a normalized range [0,1].
- Approximately linear around a central value I_0 , and non-linear outside this region.
- Enhances contrast in a selected intensity range while compressing others.

$$B(i, j) = B_{\min} + \frac{B_{\max} - B_{\min}}{1 + e^{-\frac{A(i, j) - I_0}{s}}}$$

- Parameters: $B_{\min}$, $B_{\max}$ (output range), I_0 (central value), s (slope).



Intensity transformations: Intensity mapping

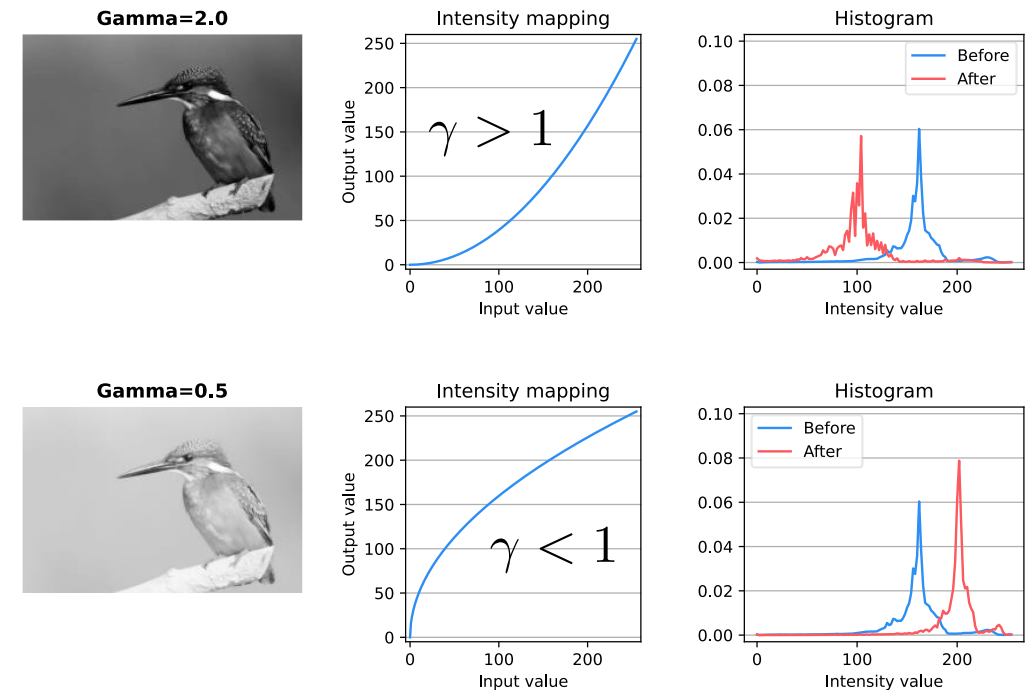
- Various functional relationships can be defined between input and output intensity values:

- **Gamma correction:**

- Applies a non-linear power-law transformation to intensity values.
- Often used for the correction of the intrinsic nonlinearity in devices or the human perception.

$$B(i, j) = A(i, j)^\gamma$$

- Parameter γ controls the shape of the mapping
 - $\gamma < 1 \rightarrow$ brightens the image
 - $\gamma > 1 \rightarrow$ darkens the image

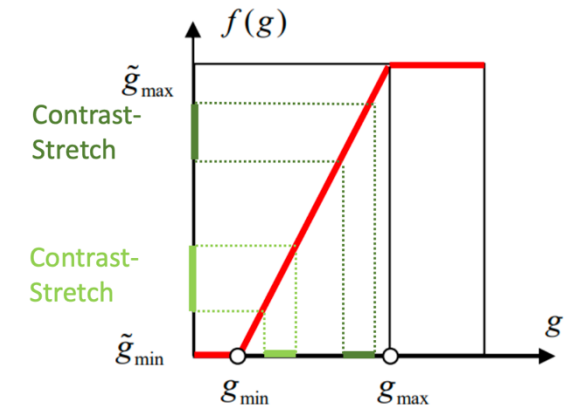


Intensity transformations: Intensity mapping

Linear mappings affect all intensities equally, while non-linear mappings enable selective contrast enhancement.

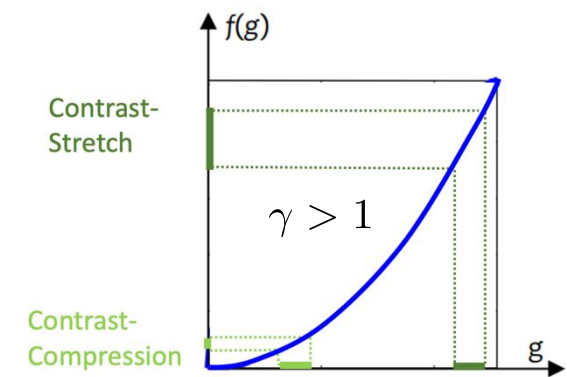
■ Linear histogram stretching (contrast stretching):

- A linear mapping expands a limited range of gray values to the full intensity range.
- Increases contrast uniformly across all intensities



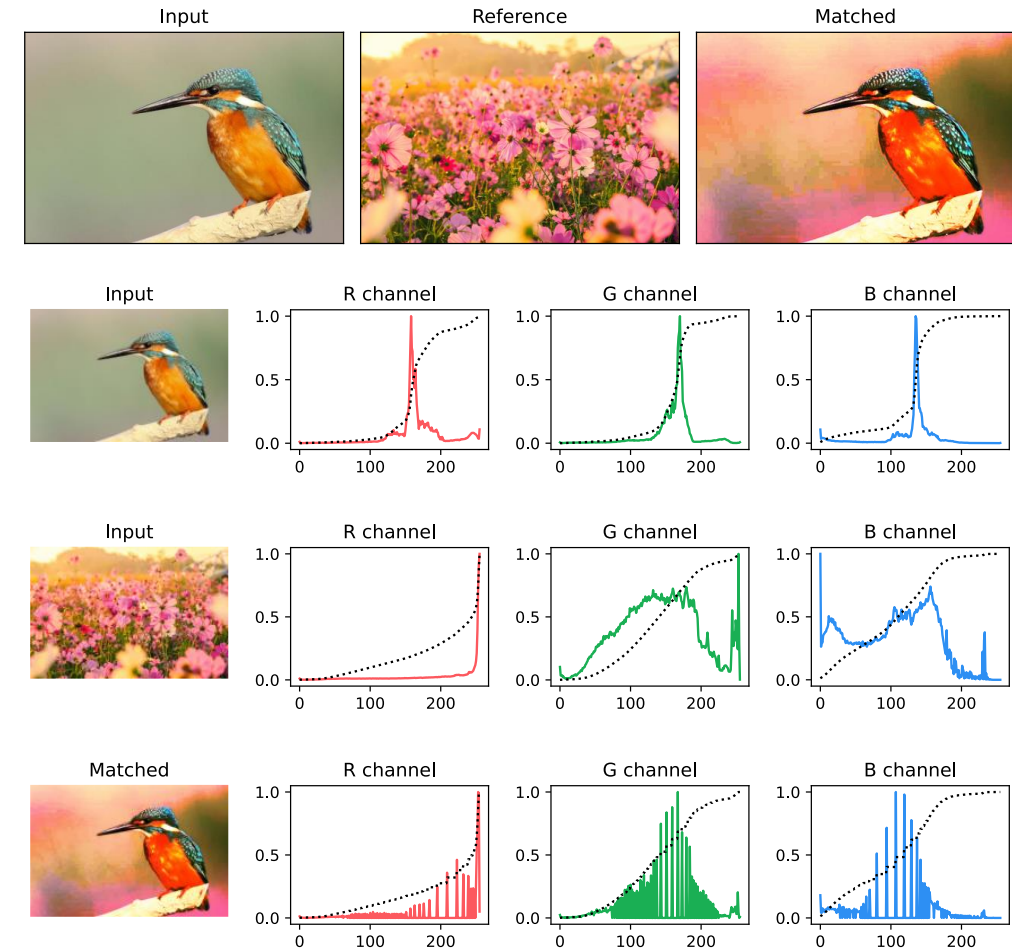
■ Gamma correction:

- A non-linear mapping that redistributes intensities unevenly.
- $\gamma < 1$: Enhances dark regions while compressing bright regions
- $\gamma > 1$: Compresses dark regions, while enhancing bright regions



Intensity transformations: Histogram-based methods

- How are intensity transformations applied?
- **Approach 2: Via intensity histograms**
 - Intensity histograms visualize the distribution (frequency) of pixel values.
 - Key idea: Use this distribution to guide the transformation!
 - Goal: Redistribute intensities to enhance contrast or achieve a more favorable intensity distribution
 - Examples:
 - Histogram equalization
 - Histogram matching
 - Adaptive methods (e.g., CLAHE)

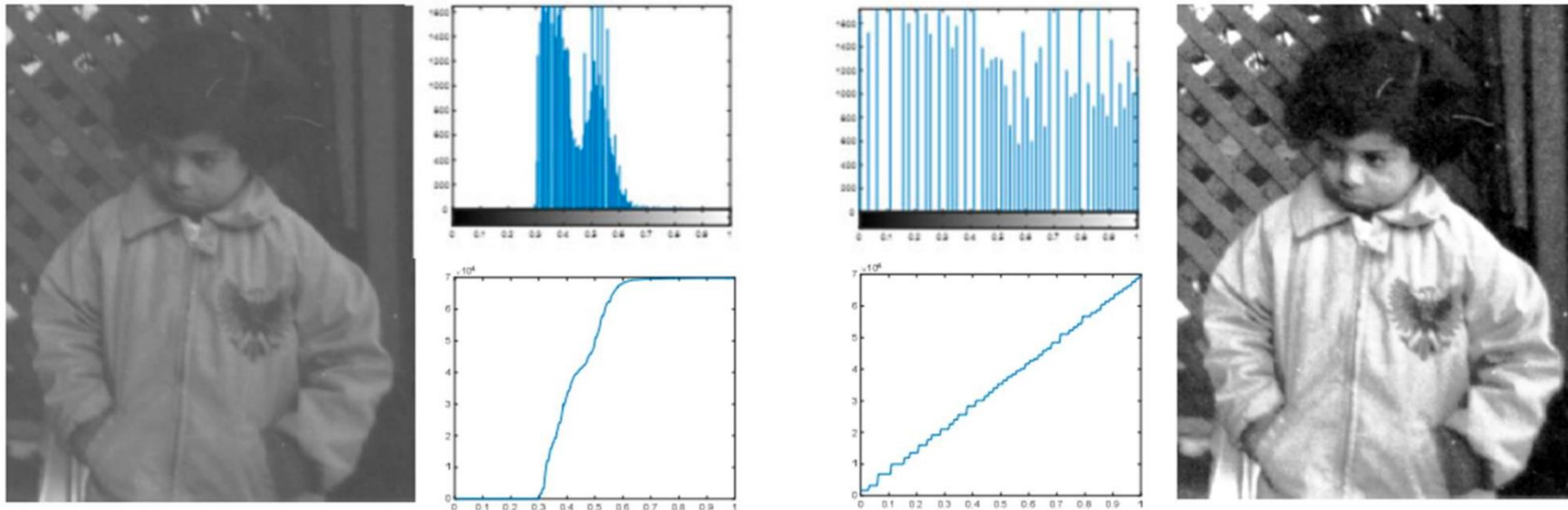


Example: An input image is transformed to match a reference image by adjusting its intensity distributions so that each channel follows the corresponding distribution of the reference.

Intensity transformations: Histogram-based methods

■ Histogram equalization:

- Idea: Redistribute gray values to achieve a more uniform intensity distribution.
- Improves the use of the available intensity range
- Enhances contrast, especially in low-contrast regions



Patch-based transformations



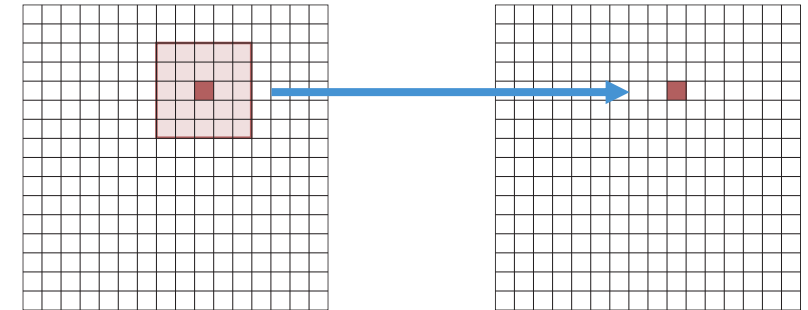
Convolution / spatial filtering...
Will be covered elsewhere!

Geometric transformations

Geometric transformations

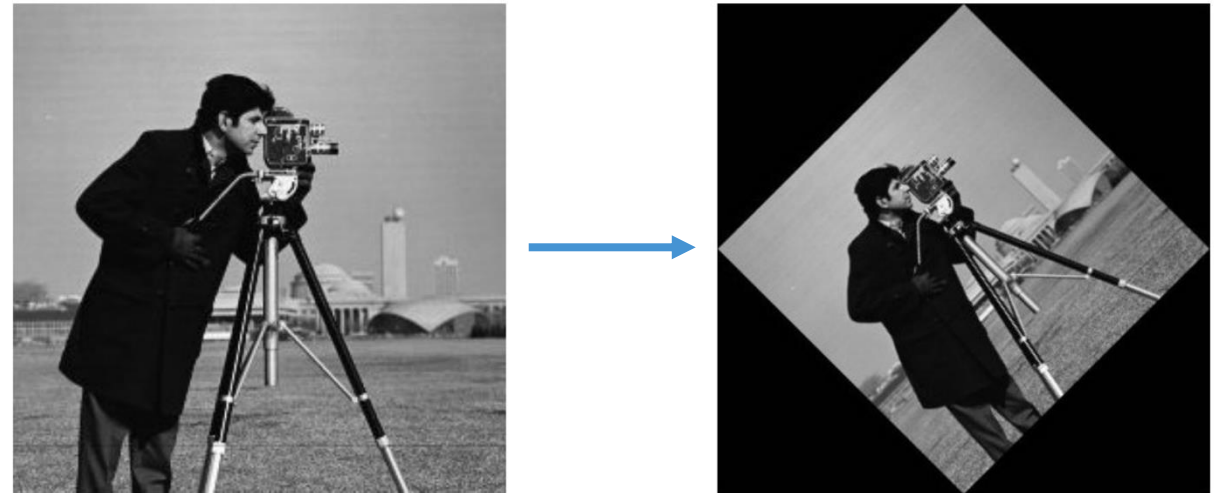
- Goal: Transform an image by modifying its spatial arrangement using:

- Translation
- Rotation
- Scaling
- (Shearing)
- (Perspective distortion)



- Reading:

- Primer on affine transformations.
- See Jupyter notebook on affine transformations



Frequency domain transformations



DFT/FFT, spectral filtering...
Will be covered elsewhere!